II.2.2. Bowen's Disease

Bowen's disease (BD) is an intraepithelial squamous cell carcinoma (SCC). The prevalence and incidence of this disease in France and elsewhere are unknown. Patients most commonly affected in published cases are in their seventh decade of life, with a predominance of women (70% of cases).

Clinical Features

Clinically, the cutaneous lesion presents as a well-demarcated, discoid, erythematous-squamous plaque, often keratotic or crusted. It typically occurs on covered skin and progresses slowly. In mucosal and semi-mucosal sites, BD may appear smooth, weeping, erythroplastic (e.g., **Queyrat's erythroplasia of the penis**), or leukoplakic. Genital forms are strongly associated with **HPV infection**.

Histopathology

The epidermis shows hyperplasia with disorganized architecture and atypical keratinocytes characterized by loss of normal polarity, hyperchromatic nuclei, anisonucleosis, increased mitotic activity, and dyskeratosis. These atypical cells involve the full thickness of the epidermis but do not breach the basement membrane. Variants of BD with clear, pagetoid, or pigmented cells have been described. Immunohistochemistry may be necessary to differentiate such variants from other conditions, such as extramammary Paget disease or melanocytic lesions.

In perineal areas, differential diagnosis with **Bowenoid papulosis** (VIN 3, PIN 3) relies on clinical presentation, duration, patient history, presence of viral cytopathic effects on histology, and, in some cases, HPV typing.

Risk of Progression to Invasive SCC

The risk of progression to invasive SCC is estimated approximately from retrospective studies. It appears to be between **3% and 5%** for cutaneous BD and around **10%** for Queyrat's erythroplasia. Progression may occur after variable latency, typically marked by the development of an ulcerated nodule within the pre-existing flat lesion. Metastatic potential following progression is considered higher than that of conventional SCC.

Key Points

- **Actinic keratosis (AK)** is an epidermal dysplasia.
- AK onset is linked to chronic UV exposure.
- Prevalence of AK is estimated at **11–25%** in adults over 40 years in the northern hemisphere, increasing with age.
- AK is a precancerous lesion with **low risk of malignant transformation** and a **high rate of spontaneous regression**.
- The Working Group (WG) considers AK a clinical precursor of SCC, sharing key pathophysiological features, but since progression is not inevitable, AK and SCC should remain distinct entities.
- **Bowen's disease** is an intraepidermal (in situ) SCC, most frequently occurring on the lower limbs.
- The **prevalence of BD is unknown**.
- The **rate of progression of BD to invasive SCC has not been precisely established**.
- AK and BD can be viewed as low-grade and high-grade intraepithelial neoplasias (or SCC in situ), respectively, analogous to lesions at other anatomical sites. However, their clinical and etiological profiles are distinct enough to warrant separate classification.
- To date, **no studies have proven the clinical benefit of treating "field cancerization"** as a strategy to prevent SCC.

II.3. Invasive Cutaneous Squamous Cell Carcinomas

The terms **"infiltrating"** and **"invasive"** are synonymous when applied to SCC and refer to any tumor that crosses the epithelial basement membrane and invades the dermis, irrespective of depth.

II.3.1. Definition, Epidemiology, and Natural History

The majority of cutaneous SCCs arise in sun-exposed areas, often from a precursor lesion such as **actinic keratosis**, with which they share multiple risk factors. Incidence increases with decreasing latitude (i.e., higher UV exposure). Skin phototype, a genetically determined factor, also influences SCC risk.

Several **co-carcinogens** have been identified, some site-specific: - **Tobacco use** – associated with SCC of the lower lip. - **Genital or anal HPV infection** – linked to anogenital SCC.

Less commonly, SCC may develop from: - **Bowen's disease** - **Chronic ulcers** - **Scars** - **Chronic inflammation** (e.g., hidradenitis suppurativa, radiodermatitis) - **De novo**, without a known precursor